ABDULA ADIL

AI Engineer | LLM · Voice AI · Computer Vision | Full Stack (.NET · Python · React Native)

Abu Dhabi, UAE • +971 50 736 5343 • adduadil38@gmail.com • linkedin.com/in/abdulaadil • abdula-adil.vercel.app

UAE Employment Visa • 30-Day Notice • MoHESR-Recognized Degree • UAE Driving License • Open to Relocate

PROFESSIONAL SUMMARY

AI Engineer building **LLM-powered enterprise systems** for the industrial and asset-integrity sector. Currently leading AI features on **VAIL-Plant** — real-time **voice chat** with the AI assistant (**Whisper** speech-to-text + **Kokoro** text-to-speech) and an **AI-vision pipeline that converts P&ID drawings into 3D models** — while serving as the **sole engineer building AI capabilities for the VAIL ERP platform**. Builds Python **FastAPI** AI microservices (Ollama, Semantic Kernel, vLLM) integrated with ASP.NET Core backends, and ships web and mobile front-ends in Angular and React Native. Deep daily practice in agentic AI engineering (Claude Code, custom agent skills, context/token optimization). Trusted with international client delivery (Sonatrach, Algeria); **3× Employee of the Month**.

AI & LLM ENGINEERING

LLM Integration: Ollama, Microsoft Semantic Kernel, vLLM (inference throughput), RAG-style retrieval, NL→SQL over enterprise data, prompt engineering, fine-tuning

Voice AI: Whisper (speech-to-text), Kokoro (text-to-speech), voice-driven conversational assistants for web and mobile

Computer Vision: AI-vision interpretation of engineering drawings (P&ID), Python post-processing for accuracy, 2D drawing → 3D model generation

AI Microservices: Python FastAPI — async endpoints, Pydantic validation, Dockerized deployment, REST integration with ASP.NET Core

Agentic Engineering: Claude Code, custom reusable agent skills, context/token optimization, AI-accelerated delivery pipelines

Edge & Industrial AI (Research): offline on-premise inference, low-latency small-model classification, System 1 / System 2 routing, SCADA (Modbus / OPC UA)

SOFTWARE ENGINEERING SKILLS

Backend: ASP.NET Core Web API, C#, Python, FastAPI, PHP / Laravel, Entity Framework Core, CQRS + MediatR, Clean Architecture, SignalR, JWT, REST APIs, microservices

Frontend & Mobile: React Native (CLI & Expo), ReactJS + Vite, Angular, Astro, TypeScript, JavaScript (ES6+), Redux, Zustand, MobX

Data & BI: SQL Server (query tuning & optimization), MySQL, Firebase, Power BI

Cloud & DevOps: Docker, Git, GitHub, Azure (fundamentals), Vercel, production deployment support

Security & Practice: OWASP Top 10, secure coding, VAPT fundamentals; Agile / Scrum

PROFESSIONAL EXPERIENCE

Software Developer — AI & Full Stack | Velosi Asset Integrity Ltd *Jan 2025 – Present | Abu Dhabi, UAE*

VAIL-Plant — Enterprise AI Assistant & Mobile App (current focus) · *React Native CLI · Angular · ASP.NET Core · FastAPI · Whisper · Kokoro · Ollama · vLLM*

- Integrated **voice chat** into the AI assistant — **Whisper** for speech-to-text and **Kokoro** for text-to-speech — so field engineers can ask questions about plant data and hear answers hands-free.
- Built an **AI-vision pipeline that converts P&ID (piping & instrumentation) drawings into 3D models**, combining vision-model extraction with Python post-processing to improve accuracy.
- Developed the **VAIL-Plant mobile app end-to-end** in React Native CLI — asset inspection, maintenance planning, risk assessment and reporting.
- Co-developed the AI backend with the team lead: **Ollama + Semantic Kernel** for natural-language querying of plant and asset data, **vLLM** for inference throughput, and conversational front-ends in Angular (web) and React Native (mobile).

VAIL ERP — AI Development (solo) · *ASP.NET Core · Python FastAPI · SQL Server · Ollama · Semantic Kernel*

- **Sole engineer** designing and building AI capabilities from scratch for a full-scale ERP (Inventory, Sales, HR, Finance) — natural-language access to enterprise data, owning architecture through delivery.
- Built **Python FastAPI microservices** for LLM inference, retrieval and NL querying, Dockerized and integrated with the ASP.NET Core backend over REST so AI services scale and deploy independently of the core ERP.
- Applied Clean Architecture with **CQRS + MediatR** for transactional .NET modules and lightweight FastAPI services for Python-native AI components.

Across Projects

- Delivered **VAIL-AIPS** process-safety modules — hazard management, compliance tracking, dashboards and risk tracking (React Native CLI · ASP.NET Core · MySQL).
- Optimized SQL Server queries to reduce response times; supported production deployments and cross-stack debugging.
- Accelerated delivery with **agentic AI workflows** — Claude Code, disciplined prompt engineering, context/token optimization and custom reusable agent skills.
- **International client delivery — Sonatrach, Algeria:** ran a 3-day onsite hands-on training program for engineering teams — live demos, real-time technical support and adoption documentation.
- **Employee of the Month ×3** — Sep 2025, Feb 2026, May 2026.

Web Development Intern | Almasat Information Technology *Feb 2024 – Apr 2024 | Dubai, Remote*

- Developed and optimized web content platforms, improving SEO performance and search visibility across client web properties.

Vulnerability Analyst Intern | Red Team Hackers *Aug 2023 – Sep 2023 | Kochi, India*

- Conducted a full penetration test (DC-1) and delivered a structured vulnerability report with remediation recommendations.

INDEPENDENT RESEARCH

The Industrial Reflex — Laya Edge AI Architecture for Millisecond Safety Decisions *Ongoing | Demos on LinkedIn*

- Two-tier, **fully offline** edge AI for offshore and industrial sites: a small on-site reflex model triages every field report and alarm in ~35 ms (equipment, possible gas release with confidence, permit-to-work, urgency), and only complex questions reach a **local LLM grounded in site procedures**.
- SCADA telemetry over **Modbus / OPC UA** → local protocol translation (<2 ms) → parallel inference → shutdown-valve decision in **<35 ms in simulation**, versus ~8.5 s typical operator response.
- Human-in-the-loop by design — guidance is advisory; people make the final decisions.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering, Computer Science — P.A. College of Engineering, Mangalore (VTU), India *2024*

Minors: Software Design, Networking • Recognized by the UAE Ministry of Higher Education and Scientific Research (MoHESR)

ASP.NET Core 8 Web API — Clean Architecture + Azure Services (Udemy) • Pre-Security Learning Path (TryHackMe)

ADDITIONAL INFORMATION

Languages: English (Professional), Malayalam (Native) **Strengths:** problem solving & debugging, system architecture, technical documentation, client training & communication, applied AI research